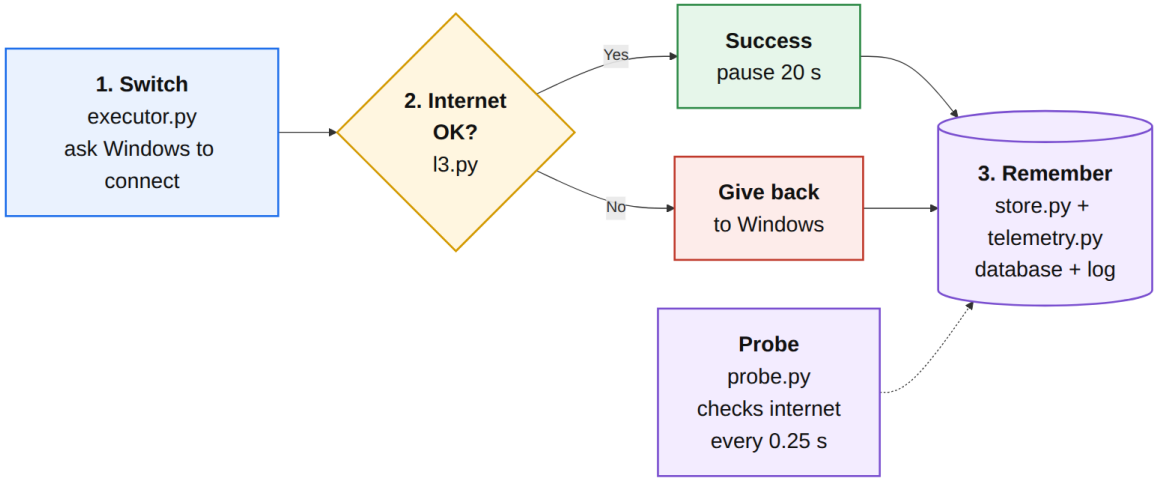
Part 3: Act and Remember

RoamPredict · short report · Adil

What Part 3 does

Parts 1 and 2 decide **when** to change access point and **which one**. Part 3 **does the switch** and **measures the result**.



Part 3 in one picture.

1. Act: switch safely

1. **Check** the new access point with a fresh scan: still there, still 8 dB stronger.
2. **Connect** with `WlanConnect`, giving only the **name** of the saved Wi-Fi profile. The password stays in Windows.
3. **Check the internet**: does the laptop have a real IP address and gateway?
4. **If something fails**: give control back to Windows. After 2 failures, or a login problem, everything stops (kill switch).

2. Remember: measure and record

- **Probe**: checks the internet every 0.25 s (TCP to Google). 2 missed answers = an outage.
- **Database** (SQLite): saves every switch and every outage, with the mode (with or without RoamPredict).
- **Log file**: writes every event, like a black box. Passwords are removed.

My files

File	Job
roam/executor.py	Does the switch: check, connect, verify, fallback
net/l3.py	IP address, gateway, ARP flush, DHCP renew
net/probe.py, net/latency.py	Internet check every 0.25 s
topology/store.py	The database
telemetry.py	The log file
platform/win32_structs.py	Windows C structures, so Python can call Windows

Example of a dictionary

```
session_outages = {"wifi": [], "internet": []}   # engine.py:200
```

One list of outage durations per type. At the end, we count, add up and take the longest: that is the STOP line.

Field testing: the data we collected

We tested RoamPredict on the real campus network: **about 10 walks over 3 days** (library, buvette, atelier), on **2 laptops**, with and without RoamPredict acting. Every walk was recorded by Part 3: each outage, each switch, each event in the log.

Problem found thanks to the recorded walks	What we changed
The Wi-Fi driver gave a nonsense signal value (+27 dBm)	Use the signal quality instead
The signal reading froze while the link was dead	New rule: if the internet is dead, trust the last scan
A crash on large Windows error codes	Fixed the error formatting
A fake signal value just after each reconnection	Ignore the trend for 9 s after reconnecting
One switch connected but got no IP address (169.254.x)	Give control back to Windows

Conclusion: collecting data on real walks made the code much more reliable: every problem above was found in the database and the log, fixed, and checked by an automated test. The data also showed how much Windows alone varies (45% to 92% of a walk offline on the same route). To prove that RoamPredict is better in general, the next step is more walks on the same route, with and without it.